UE23AM343BB4 – Machine learning for Finance Course Project

Project Title : Temporal and Explainable Graph Neural Network for Illicit Transaction Detection in Bitcoin Networks

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Outline

- Introduction about the Problem domain
- Problem statement
- Literature survey: (Base paper discussion)
- Research Gap
- Dataset Description
- Proposed methodology/Approach (Block diagram/ Pipeline)
- Explain each Block in the block diagram (pipeline of the work)
- Results and Discussion
- Conclusion and Future scope
- References

Introduction

- Bitcoin transactions are **pseudo-anonymous**, making fraud detection difficult
- Illicit activities include:
  - Money laundering
  - Dark web transactions
- Traditional ML fails because:
  - Ignores relationships between transactions
- Bitcoin network forms a **graph structure**
- Graph Neural Networks (GNNs) can capture:
  - Transaction dependencies
  - Hidden fraud patterns

Problem statement

The rapid growth of cryptocurrency transactions, particularly in Bitcoin networks, has led to an increase in illicit activities such as money laundering, fraud, and illegal trading. Due to the pseudo-anonymous nature of Bitcoin, identifying such illicit transactions is a significant challenge for financial institutions and regulatory authorities.

Traditional machine learning models treat transactions as independent data points and fail to capture the complex relationships between transactions in a network. Additionally, the highly imbalanced nature of transaction data makes it difficult to accurately detect rare illicit activities.

This project aims to develop a robust machine learning framework that leverages Graph Neural Networks (GNNs) to model the transactional network structure and accurately classify transactions as licit or illicit. The proposed system also addresses class imbalance and aims to achieve high detection rates while minimizing false negatives, thereby enabling effective and scalable anti-money laundering (AML) solutions.

Literature survey - Base Paper

Anti-Money Laundering in Bitcoin (Weber et al., 2019)

- Introduced the **Elliptic Bitcoin Dataset**, widely used for AML research
- Applied traditional machine learning models such as:
  - Logistic Regression
  - Random Forest
- Used **hand-crafted transaction features** like:
  - Transaction volume
  - Time-based features
- Treated each transaction as an **independent data point**
- Achieved moderate performance in detecting illicit transactions

Limitations:

- Does not consider **relationships between transactions**
- Fails to capture **network structure and transaction flow patterns**

Insight for our work:

- Motivates the use of **graph-based models** instead of traditional ML

Literature survey

Graph Convolutional Networks (Kipf & Welling, 2017)

- Introduced **Graph Convolutional Networks (GCN)** for node classification
- Designed to work on **graph-structured data**
- Each node updates its representation by aggregating:
  - Its own features
  - Features of neighboring nodes

Key Concept:

- Nodes connected in a graph tend to have **similar properties**

Advantages:

- Captures **local graph structure**
- Improves classification performance compared to traditional ML

Limitations:

- Treats all neighbors equally
- Does not handle **dynamic or time-evolving graphs**

Insight for our work:

- Forms the **core model** for learning transaction relationships

Literature survey

Graph Attention Networks (Veličković et al., 2018)

- Extends GCN using an **attention mechanism**
- Assigns different importance (weights) to neighboring nodes
- Learns which neighbors are more relevant for prediction

Key Concept:

- Not all transactions influence each other equally

Advantages:

- More flexible than GCN
- Better representation of complex relationships

Limitations:

- Computationally expensive
- Harder to scale for large graphs

Insight for our work:

- Highlights the importance of **weighted relationships in transaction networks**
- Suggests future improvement over GCN

Literature survey

Temporal Graph Neural Networks

- Extends GNNs to handle **time-dependent graphs**
- Models how relationships between nodes evolve over time
- Uses time-stamped transaction data

Key Concept:

- Fraud patterns change over time
- Static models may miss evolving behaviors

Advantages:

- Captures **dynamic fraud patterns**
- More realistic for financial systems

Limitations:

- Complex architecture
- Higher computational cost

Insight for our work:

- Current model is static
- Temporal modeling is a **future extension**

Literature survey

Explainability in Graph Neural Networks (GNNExplainer)

- Provides methods to **interpret GNN predictions**
- Identifies:
  - Important features
  - Important subgraphs influencing decisions

Key Concept:

- Understanding *why* a transaction is flagged is critical

Advantages:

- Improves **trust and transparency**
- Important for financial regulations

Limitations:

- Adds computational overhead
- Interpretation may not always be straightforward

Insight for our work:

- Adds value for **real-world deployment**
- Included as part of future work

Research Gap

Existing methods have several limitations:

- Treat graph as static and ignore time evolution
- Do not combine feature-based and graph-based learning
- Lack interpretability for financial applications

Limited focus on:

- Real-world fraud detection systems
- Risk scoring and alert generation

Need for a solution that:

- Integrates multiple learning approaches
- Handles class imbalance effectively
- Provides practical deployment capability

Dataset Description

Dataset used: **Elliptic Bitcoin Dataset**

Contains:

- ~46,000 labeled transactions
- 49 features per transaction
- Graph structure of transaction links

Features include:

- Transaction volume
- Time-related features
- Graph-based features

Labels:

- 0 → Licit
- 1 → Illicit

Dataset characteristics:

- Highly imbalanced
- Real-world financial data
- Includes temporal transaction batches

Proposed Methodology/Block Diagram

Proposed system follows a structured pipeline:

1. Data preprocessing and cleaning
2. Graph construction from transaction data
3. Hybrid model combining MLP and GCN
4. Model training and evaluation
5. Fraud detection simulation

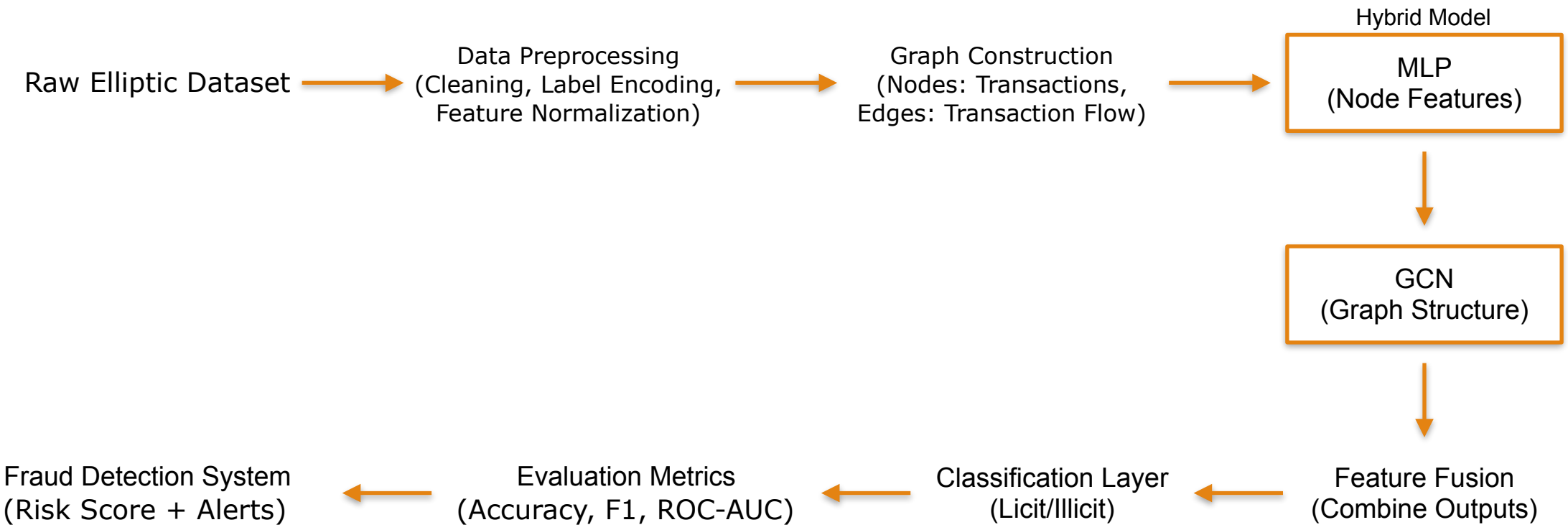
Key idea:

- Combine:
  - Node-level feature learning
  - Graph-based relationship learning

Expected outcome:

- Improved detection of illicit transactions

Proposed Methodology/Block Diagram



Data Preprocessing

Merge transaction features with class labels

Remove transactions with unknown labels

Encode labels into binary format (0 and 1)

Apply feature normalization:

- StandardScaler ensures stable training

Handle class imbalance using:

- Weighted loss function

Output:

- Clean, normalized dataset ready for modeling

Graph Construction

- Convert dataset into graph representation:
  - Nodes represent transactions
  - Edges represent transaction flow
- Create:
  - Node feature matrix
  - Edge index for graph connections
- Map transaction IDs to node indices
- Purpose:
  - Capture dependencies between transactions
  - Enable graph-based learning

Block 3: Hybrid Model

Hybrid Model

- Hybrid model combines two learning approaches:

MLP (Multi-Layer Perceptron):

- Learns patterns from transaction features

GCN (Graph Convolutional Network):

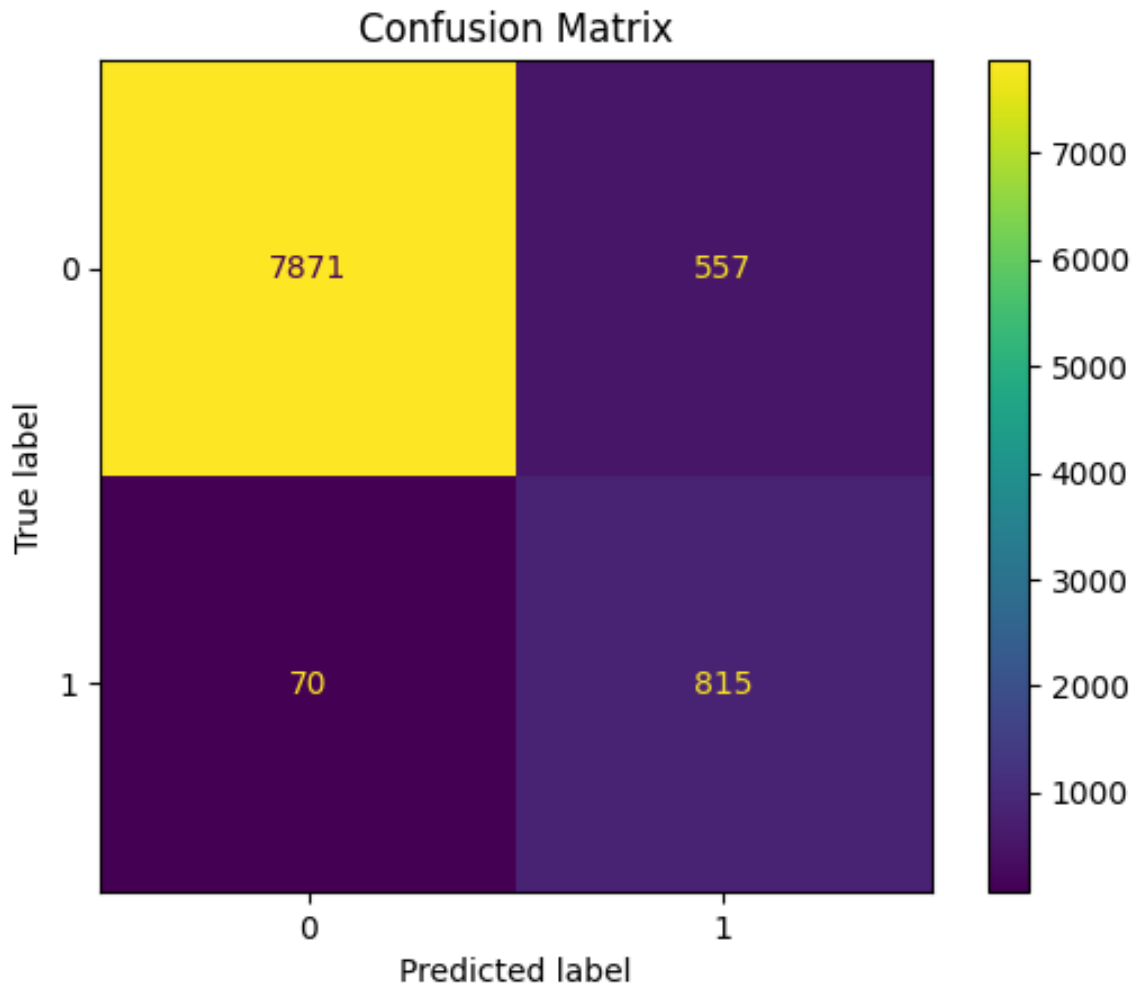
- Learns relationships between connected transactions
- Outputs from both models are combined
- Final prediction obtained using softmax
- Advantage:
  - Captures both local and global information
  - Improves classification performance

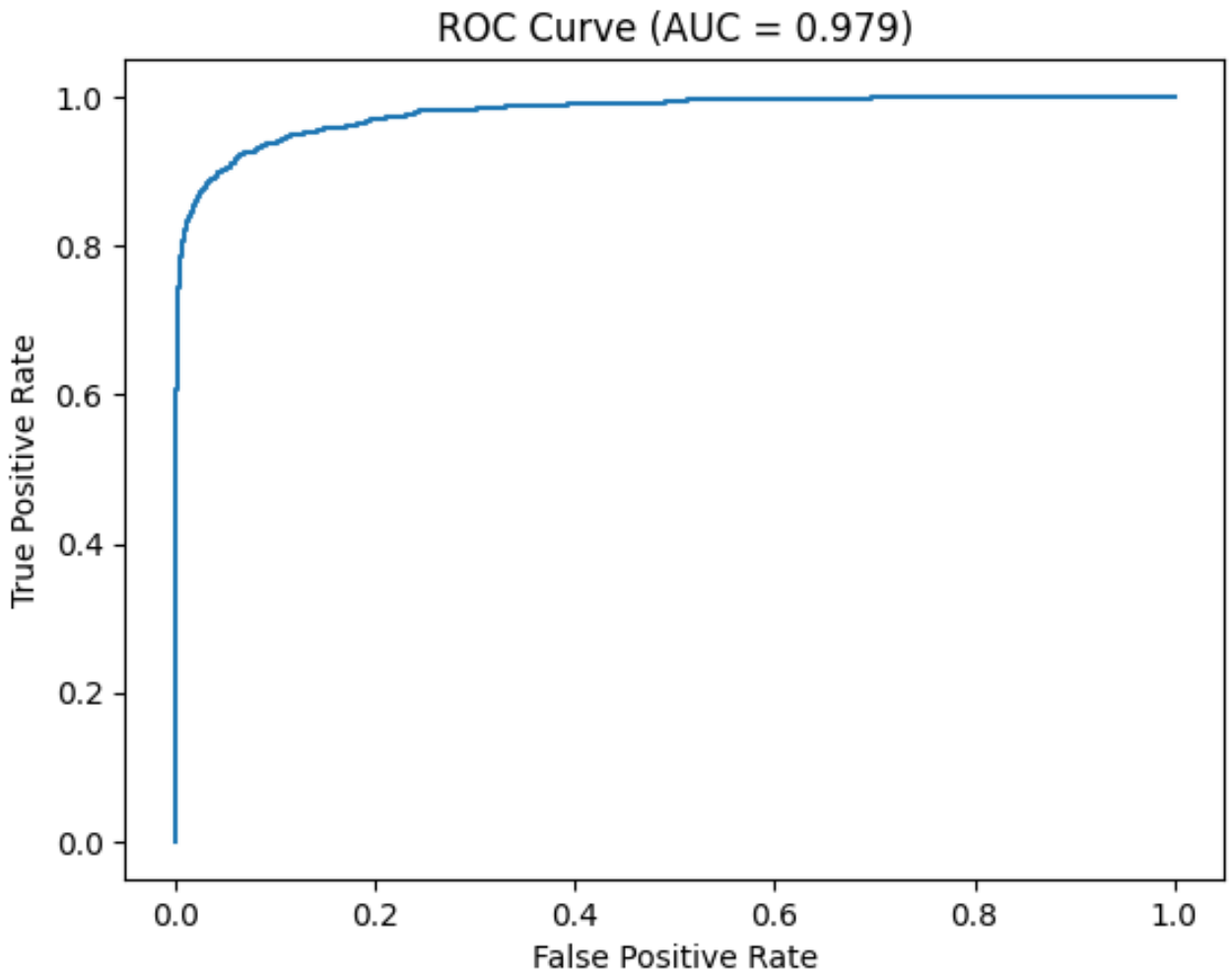
Results and Discussion

```
Starting Bitcoin AML GNN Project
Loading dataset...
Total labeled transactions: 46564
Total nodes: 46564
Total edges: 36624
Nodes: 46564, Edges: 36624
Epoch 001 | Loss: 1.1662 | Val Acc: 0.4932
Epoch 002 | Loss: 0.6374 | Val Acc: 0.6451
Epoch 003 | Loss: 0.4601 | Val Acc: 0.7404
Epoch 004 | Loss: 0.4066 | Val Acc: 0.7975
Epoch 005 | Loss: 0.3841 | Val Acc: 0.8297
Epoch 006 | Loss: 0.3775 | Val Acc: 0.8469
Epoch 007 | Loss: 0.3744 | Val Acc: 0.8503
Epoch 008 | Loss: 0.3661 | Val Acc: 0.8447
Epoch 009 | Loss: 0.3531 | Val Acc: 0.8381
Epoch 010 | Loss: 0.3411 | Val Acc: 0.8289
Epoch 011 | Loss: 0.3292 | Val Acc: 0.8261
Epoch 012 | Loss: 0.3203 | Val Acc: 0.8295
Epoch 013 | Loss: 0.3113 | Val Acc: 0.8407
Epoch 014 | Loss: 0.2991 | Val Acc: 0.8566
Epoch 015 | Loss: 0.2911 | Val Acc: 0.8697
Epoch 016 | Loss: 0.2830 | Val Acc: 0.8806
Epoch 017 | Loss: 0.2764 | Val Acc: 0.8832
Epoch 018 | Loss: 0.2675 | Val Acc: 0.8853
Epoch 019 | Loss: 0.2614 | Val Acc: 0.8798
Epoch 020 | Loss: 0.2536 | Val Acc: 0.8742
Epoch 021 | Loss: 0.2500 | Val Acc: 0.8725
Epoch 022 | Loss: 0.2453 | Val Acc: 0.8750
Epoch 023 | Loss: 0.2409 | Val Acc: 0.8849
Epoch 024 | Loss: 0.2360 | Val Acc: 0.8916
Epoch 025 | Loss: 0.2284 | Val Acc: 0.8986
Epoch 026 | Loss: 0.2268 | Val Acc: 0.9047
Epoch 027 | Loss: 0.2221 | Val Acc: 0.9064
```

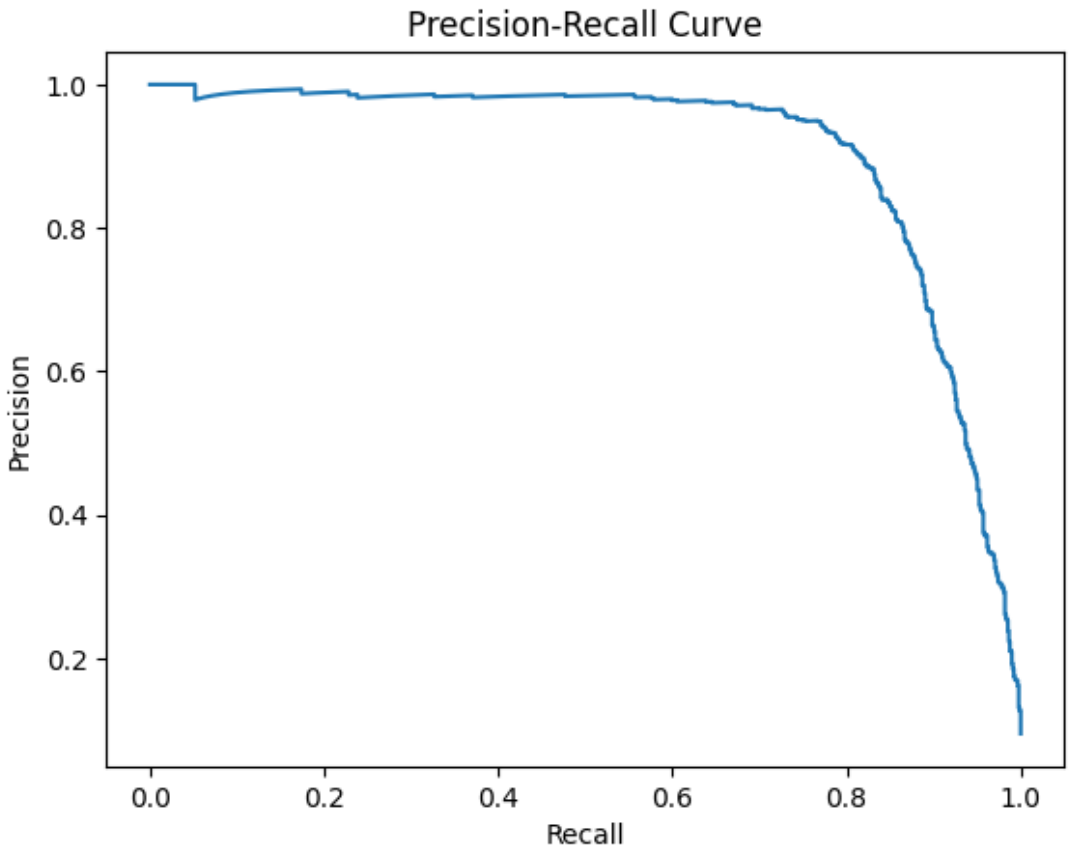
```
Epoch 028 | Loss: 0.2158 | Val Acc: 0.9066
Epoch 029 | Loss: 0.2136 | Val Acc: 0.9047
Epoch 030 | Loss: 0.2104 | Val Acc: 0.9059
Epoch 031 | Loss: 0.2070 | Val Acc: 0.9079
Epoch 032 | Loss: 0.2055 | Val Acc: 0.9109
Epoch 033 | Loss: 0.1991 | Val Acc: 0.9143
Epoch 034 | Loss: 0.1961 | Val Acc: 0.9171
Epoch 035 | Loss: 0.1945 | Val Acc: 0.9171
Epoch 036 | Loss: 0.1917 | Val Acc: 0.9163
Epoch 037 | Loss: 0.1890 | Val Acc: 0.9156
Epoch 038 | Loss: 0.1856 | Val Acc: 0.9148
Epoch 039 | Loss: 0.1826 | Val Acc: 0.9167
Epoch 040 | Loss: 0.1834 | Val Acc: 0.9175
Epoch 041 | Loss: 0.1789 | Val Acc: 0.9188
Epoch 042 | Loss: 0.1761 | Val Acc: 0.9199
Epoch 043 | Loss: 0.1760 | Val Acc: 0.9223
Epoch 044 | Loss: 0.1725 | Val Acc: 0.9221
Epoch 045 | Loss: 0.1705 | Val Acc: 0.9227
Epoch 046 | Loss: 0.1685 | Val Acc: 0.9253
Epoch 047 | Loss: 0.1665 | Val Acc: 0.9279
Epoch 048 | Loss: 0.1643 | Val Acc: 0.9285
Epoch 049 | Loss: 0.1651 | Val Acc: 0.9281
Epoch 050 | Loss: 0.1617 | Val Acc: 0.9281

Training Complete

Test Set Evaluation:
              precision    recall  f1-score   support

           0       0.99      0.93      0.96      8428
           1       0.59      0.92      0.72       885

    accuracy                           0.93      9313
   macro avg       0.79      0.93      0.84      9313
weighted avg       0.95      0.93      0.94      9313

ROC-AUC: 0.9787855386537745
F1 Score: 0.7221976074435091

Running Fraud Detection Simulation...
Flagged Transactions: 3726
```

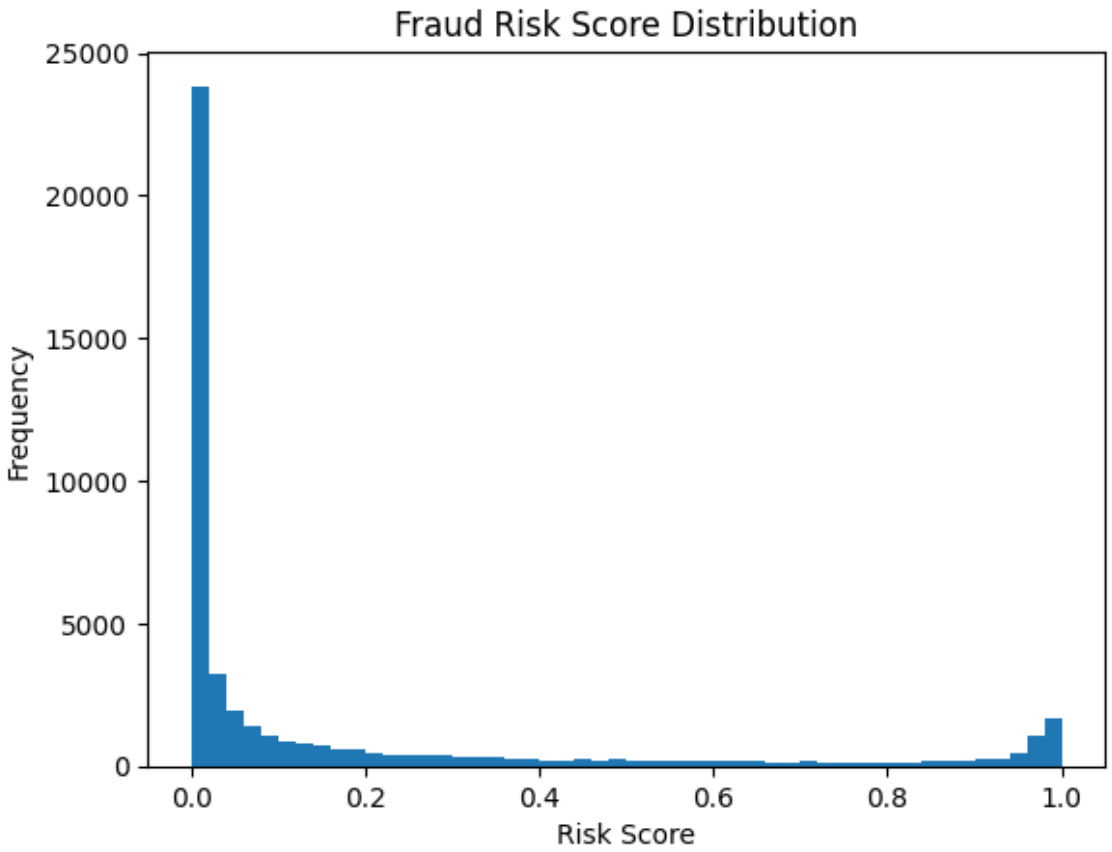
Results and Discussion



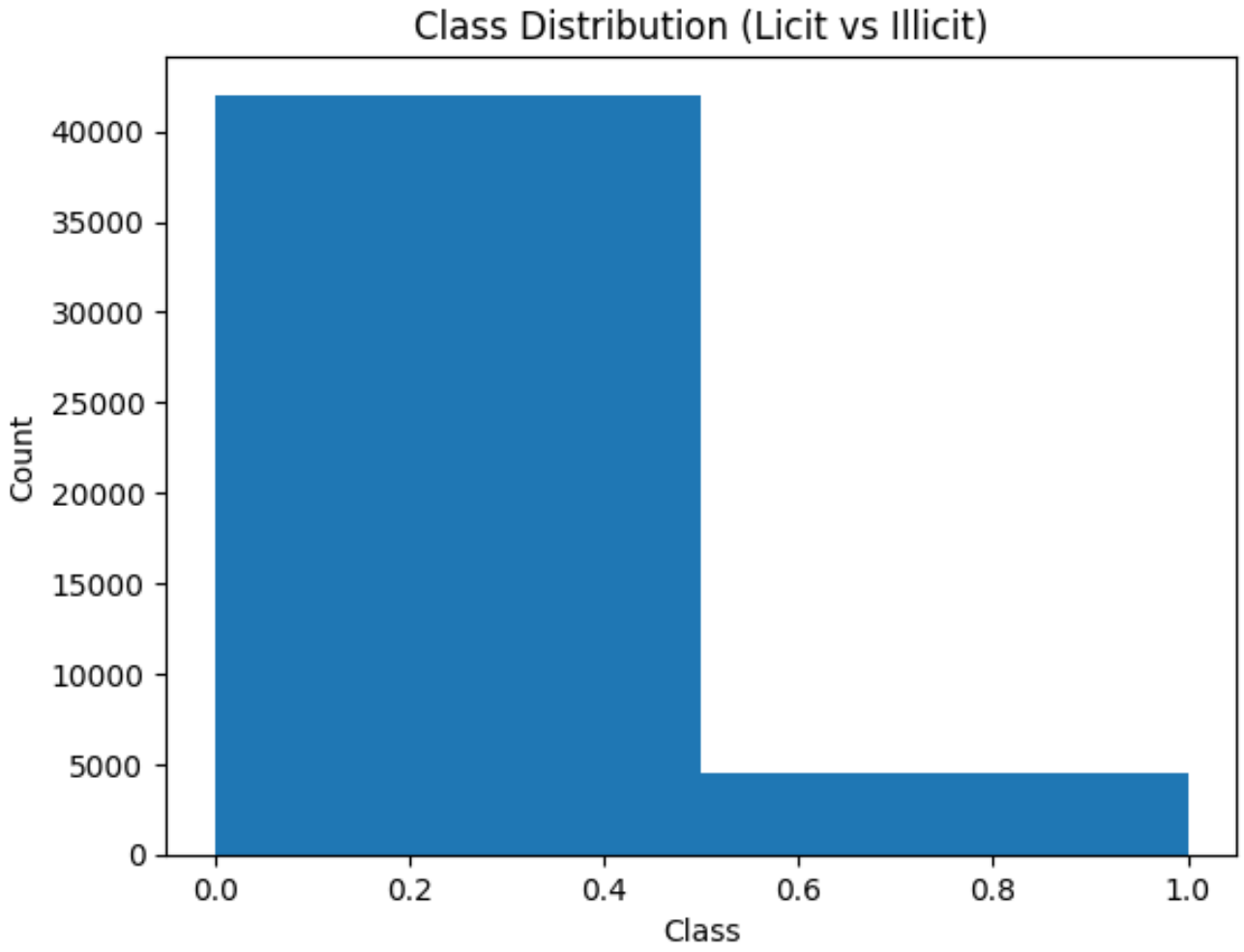


Results and Discussion





Results and Discussion




```
===== MODEL PERFORMANCE METRICS =====

Accuracy  : 0.9327
Precision : 0.5940
Recall    : 0.9209
F1 Score  : 0.7222

===== CLASSIFICATION REPORT =====

              precision    recall  f1-score   support

           0       0.99      0.93      0.96      8428
           1       0.59      0.92      0.72       885

    accuracy                           0.93      9313
   macro avg       0.79      0.93      0.84      9313
weighted avg       0.95      0.93      0.94      9313


ROC-AUC   : 0.9788

===== CONFUSION MATRIX =====
[[7871  557]
 [  70  815]]
```

Results and Discussion

Model performance:

- Accuracy: **93%**
- ROC-AUC: **0.978**
- F1 Score: **0.72**

Illicit transaction detection:

- Recall: **92%** (high detection rate)
- Precision: **59%**

Observations:

- Model successfully identifies most fraud cases
- Some normal transactions are misclassified

Trade-off:

- Higher recall is preferred in fraud detection
- Missing fraud is more costly than false alerts

Conclusion and Future Scope

Conclusion:

- Graph-based learning improves fraud detection significantly
- Hybrid model effectively combines feature and graph information
- Model performs well on real-world financial data

Future Scope:

- Incorporate temporal graph models
- Use Graph Attention Networks for better performance
- Add explainability techniques (SHAP, GNNExplainer)
- Deploy system for real-time fraud detection

References

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Thank You